

## HEALTHCARE ANALYTICS

Portfolio Case Study

# Healthcare KPI Performance Review

## *Demo Case Study*

A portfolio case study demonstrating end-to-end KPI reporting: quarterly tracking, regional deep-dives, executive-ready summaries, and root-cause analysis.

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Data Analysis · KPI Reporting · Performance Dashboards | 2026

*All KPI values, region names, organizational details, and visuals in this case study are fully fictional, created solely for portfolio demonstration purposes. No confidential organizational data is displayed.*

# About This Demonstration



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## Division Scope & Role

This demo simulates a healthcare performance reporting function responsible for tracking service-line KPIs across multiple departments and regional units. The role covers performance monitoring, standardized quarterly reporting cycles, and translating operational data into executive-ready insight.



## What This Case Study Shows

- Structuring raw KPI data into a performance dashboard
- Strategic KPI tracking across quarterly cycles
- Regional deep-dives with monthly RAG monitoring
- Root-cause analysis and action-planning for underperforming KPIs
- Department-level KPI scorecards

# KPI Categories & Reporting Scope

The reporting function covers three KPI tiers, monitored monthly and consolidated into a quarterly executive review.

8

## Strategic KPIs

Measure achievement of long-term strategic objectives and division-level goals.

14

## Executive KPIs

Measure the performance of each department against its operating targets.

62

## Regional KPIs

Measure regional-unit performance, tracked monthly with RAG status per region.

**84 KPIs total** · reported across 6 departments and 12 regional units · monthly data collection, quarterly executive consolidation

# Overall Performance Dashboard

TOTAL KPIs REPORTED

**24**

Across all departments

ON / ABOVE TARGET

**87.5%**

21 of 24 KPIs achieving

BELOW TARGET

**12.5%**

3 of 24 KPIs at risk

WEIGHTED ACHIEVEMENT

**106.3%**

Trend: improving ↗



## Key Highlights

- 21 of 24 KPIs on or above target (87.5%) — strong performance this reporting cycle.
- OBJ-01 and OBJ-02 exceeded 110%+ — strong progress in quality governance and access.
- KPI-03, KPI-07, and KPI-11 achieved full 12/12 regional ratings across all units.



## Focus Areas

- OR Utilization: only 4/12 regions meet target — scheduling efficiency gaps need addressing.
- Preventive Screening Coverage: 38% vs. 50% target — reporting activation required.
- ED 4-Hour Discharge: 3/12 regions below target — patient-flow bottlenecks under review.

# Quarterly KPI Tracking

Strategic KPIs are tracked quarter over quarter against fixed annual targets. Sample data shown for Q1–Q3.

| KPI                                     | Q1 Actual | Q1 Target | Ach%   | Q2 Actual | Q2 Target | Ach%   | Q3 Actual | Q3 Target | Ach%   |
|-----------------------------------------|-----------|-----------|--------|-----------|-----------|--------|-----------|-----------|--------|
| Access to essential health services (%) | 95.8%     | 95.0%     | 100.8% | 96.1%     | 95.0%     | 101.2% | 96.4%     | 95.0%     | 101.5% |
| ED 4-hour discharge rate (%)            | 86%       | 90%       | 95.6%  | 89%       | 90%       | 98.9%  | 91%       | 90%       | 101.1% |
| Average length of stay (days)           | 4.7       | 4.5       | 95.7%  | 4.5       | 4.5       | 100.0% | 4.4       | 4.5       | 102.3% |
| Patient experience score                | 83.2      | 82.0      | 101.5% | 84.0      | 82.5      | 101.8% | 85.1      | 83.0      | 102.5% |
| Preventive screening coverage (%)       | 31%       | 50%       | 62.0%  | 35%       | 50%       | 70.0%  | 38%       | 50%       | 76.0%  |
| Critical care beds per 100K population  | 15.2      | 14.0      | 108.6% | 15.4      | 14.0      | 110.0% | 15.5      | 14.0      | 110.7% |

Ach% for lower-is-better KPIs (e.g., length of stay) is computed as  $\text{target} \div \text{actual}$ .

# Executive KPI Summary

| Objective                                  | Target | Actual | Ach. % |
|--------------------------------------------|--------|--------|--------|
| OBJ-01 Quality & Patient Safety Governance | 100%   | 112%   | 112%   |
| OBJ-02 Access & Service Coverage           | 100%   | 110%   | 110%   |
| OBJ-03 Operational Efficiency              | 100%   | 97%    | 97%    |
| OBJ-04 Data & Reporting Maturity           | 100%   | 81%    | 81%    |



**Watch item:** OBJ-04 underperformed at 81% — driven primarily by low results on the preventive screening coverage sub-metric (38% vs. a 50% target). Tighter integration between site-level reporting and the central dashboard is required to capture true activity.

# Performance Analysis



## Top Performing KPIs

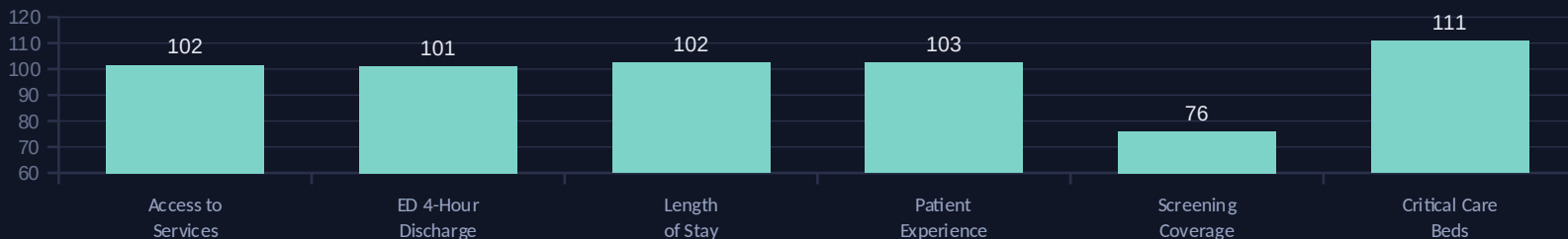
- Critical care bed capacity sustained strong, consistent performance across the cycle.
- Patient experience improved steadily — a +1.9-point gain over the start of the cycle.
- Access to essential services exceeded its benchmark every quarter.



## Bottom Performing KPIs

- Preventive screening coverage reached 38% against a 50% target — data capture gaps.
- ED 4-hour discharge remained close to target but volatile across regions.
- Length of stay, while trending favorably, remains sensitive to seasonal demand.

Achievement % by KPI (vs. 100% Target) — Q3



# Root Cause & Action Plan Example

Focus KPI: Preventive Screening Coverage (Target: 50%, Actual: 38%)



## Root Cause

No structured data-submission mechanism exists between reporting sites and the central dashboard. Manual processes are not integrated with the underlying system, so real screening activity in the field is only partially captured.



## Action Plan

Activate system integration between site-level reporting and the central dashboard. Mandate monthly submission. Set an interim milestone of 45% next cycle, building toward the full 50% target the following cycle.



## Status

In Progress — integration scoping underway with the data systems team; first monthly reporting cycle targeted for next period. Owner and deadline tracked in the departmental action log.



## Expected Impact



Improve reporting visibility



Increase KPI coverage



Support executive decisions



Improve consistency



# KPI-14 — Ward Occupancy Rate

RAG STATUS

9 / 12

Regions meeting standard

TREND



Up

vs. previous period

OVERALL

On Track

KPI status

Target: **≥ 80%** • Monthly regional values, sample cycle (Jan–Apr)

| Region          | Jan   | Feb   | Mar   | Apr   |
|-----------------|-------|-------|-------|-------|
| North Region    | 81.2% | 82.4% | 83.1% | 84%   |
| South Region    | 78.5% | 80.1% | 81.6% | 82.2% |
| East Region     | 83.4% | 84%   | 82.7% | 85.1% |
| West Region     | 76.2% | 77.8% | 79.5% | 80.6% |
| Central Region  | 84.8% | 85.3% | 86%   | 86.4% |
| Coastal Region  | 72.1% | 73.5% | 75.2% | 76.8% |
| Highland Region | 80.4% | 81%   | 80.2% | 81.9% |
| Valley Region   | 68.9% | 70.4% | 71.8% | 73%   |
| Delta Region    | 82.6% | 83.2% | 84.5% | 84.9% |
| Plateau Region  | 79%   | 80.6% | 81.1% | 82%   |
| Lakeside Region | 85.7% | 86.1% | 85.4% | 87.2% |
| Summit Region   | 74.3% | 75.9% | 77.6% | 79.1% |

Green ≥ 80% (target met) • Amber 70–79.9% • Red &lt; 70%. All values are fictional sample data.

# KPI-20 — OR Utilization

RAG STATUS

4 / 12

Regions meeting standard

TREND

→ Flat

vs. previous period

OVERALL

At Risk

KPI status

Target: ≥ 75% • Monthly regional values, sample cycle (Jan–Apr)

| Region          | Jan   | Feb   | Mar   | Apr   |
|-----------------|-------|-------|-------|-------|
| North Region    | 70.6% | 72.1% | 69.8% | 71.4% |
| South Region    | 66.3% | 65.9% | 67.2% | 68%   |
| East Region     | 78.5% | 79.2% | 77.8% | 80.1% |
| West Region     | 62.4% | 63.1% | 61.8% | 64.2% |
| Central Region  | 76.9% | 77.4% | 78%   | 78.6% |
| Coastal Region  | 68.7% | 69.3% | 70.1% | 69.5% |
| Highland Region | 58.2% | 59.6% | 61%   | 62.3% |
| Valley Region   | 75.4% | 76%   | 75.8% | 76.5% |
| Delta Region    | 64.8% | 66.2% | 65.5% | 67.1% |
| Plateau Region  | 71.9% | 70.8% | 72.4% | 73%   |
| Lakeside Region | 77.2% | 78.1% | 76.9% | 78.8% |
| Summit Region   | 60.5% | 62%   | 63.4% | 64.9% |

Green ≥ 75% (target met) • Amber 65–74.9% • Red < 65%. All values are fictional sample data.

# Sample KPI Scorecard

*Illustrative regional scorecard — fictional sample data, created for demonstration purposes.*

| Region         | KPI-14<br>(Target ≥80%) | KPI-18<br>(Target ≤5%) | KPI-22<br>(Target ≥85%) | RAG Status |
|----------------|-------------------------|------------------------|-------------------------|------------|
| North Region   | 84.0%                   | 1.4%                   | 91.2%                   | On Track   |
| South Region   | 82.2%                   | 3.6%                   | 88.4%                   | On Track   |
| East Region    | 73.0%                   | 4.8%                   | 79.5%                   | At Risk    |
| West Region    | 80.6%                   | 2.1%                   | 87.0%                   | On Track   |
| Central Region | 64.2%                   | 6.3%                   | 71.8%                   | Off Track  |
| Coastal Region | 76.8%                   | 3.9%                   | 83.1%                   | At Risk    |

**RAG legend:** On Track ≥ 80% of sub-targets met · At Risk 60–79% · Off Track < 60%. Status reflects sample data only.

# Monthly Reporting Workflow

Each reporting cycle follows a standardized flow from raw data collection to executive presentation.



**Cadence:** monthly regional tracking → quarterly executive consolidation → annual target review.

# My Contribution



Analyzed KPI datasets and identified performance trends across departments and regions



Structured executive-ready KPI reporting presentations for leadership review



Developed department scorecards, regional deep-dives, and performance summaries



Communicated insights through clear visual reporting and RAG-based tracking



Supported leadership decision-making with root-cause analysis and action planning

# Tools & Skills



## Tools



Excel



PowerPoint



Power BI



AI-assisted Analysis



## Skills

KPI Reporting

Performance Analysis

Executive Reporting

Data Visualization

Business Analysis

Insight Communication

# Key Takeaways

1

## Data tells a story when it's structured well

Raw KPI feeds become decision-ready insight through clear categorization (strategic, executive, regional) and consistent reporting templates.

2

## Executive audiences need the headline first

Leading with achievement rate, RAG status, and trend direction lets decision-makers act fast, with regional detail available for those who need it.

3

## Root-cause framing turns gaps into action

Pairing underperforming KPIs with root causes and concrete action plans is what moves a report from reporting to driving improvement.

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